

Type → Enzyme defect → Accumulated substance → Clinical picture → Diagnosis → Treatment

Porphyrias

Headings

- Definition
- Classification
- Acute Intermittent Porphyria
- Congenital Erythropoietic Porphyria
- Erythropoietic Protoporphyria
- Other important porphyrias
- Diagnosis
- Treatment

1. Definition

- **Porphyrias** are a group of **inborn errors of heme biosynthesis**.
- They are characterized by increased production and excretion of **porphyrins and/or their precursors (ALA and PBG)**.
- Most are inherited as **autosomal dominant** disorders.

2. Classification

Based on the major site of enzyme defect:

1. **Hepatic porphyrias**
2. **Erythropoietic porphyrias**
3. **Mixed hepatic + erythropoietic porphyrias**

Clinically, remember two major patterns:

- 🧠 **Neurovisceral** → abdominal pain + neurological/psychiatric symptoms
- ☀️ **Cutaneous** → photosensitivity + skin lesions

3. Acute Intermittent Porphyria (AIP)



Cause

- **Autosomal dominant**
- Deficiency of **PBG deaminase (uroporphyrinogen-I-synthase)**
- ↓ PBG deaminase → ↑ **ALA + PBG**

Clinical features

Mainly neurovisceral — NO photosensitivity

- Acute **abdominal pain** — most common
- Nausea/vomiting may occur
- **Neurological manifestations:**
 - Sensory disturbances
 - Motor weakness
 - Neuropathy
- **Psychiatric manifestations:**
 - Confusion
 - Agitation
 - Psychiatric symptoms

Important points

- Symptoms occur **intermittently**.
- Attacks may be precipitated by:
 - **Starvation**
 - Porphyrinogenic drugs, e.g. **barbiturates**
- Symptoms improve with **high-carbohydrate intake**.

- **No photosensitivity** because porphyrins are not increased.

Diagnosis

- ↑ **ALA and PBG** in blood and urine.
- Urine may become **dark/red on standing** due to oxidation of PBG.
- **Ehrlich's reagent** detects porphyrin precursors.
- Fresh urine should be protected from light for PBG estimation.

Treatment

- Remove precipitating factors/drugs.
- **High carbohydrate/glucose** during acute attacks.
- **Heme/hemin therapy** in significant acute attacks.

 **AIP = Abdominal + ALA/PBG + Psychiatric + no Photosensitivity**

4. Congenital Erythropoietic Porphyria



Cause

- **Autosomal recessive**
- Defect in **uroporphyrinogen III synthase**
- Excess formation of **type I porphyrins**

Clinical features

Mainly **cutaneous/photosensitivity**:

- Severe **photosensitivity**
- Recurrent dermatitis
- Skin ulceration and scarring
- Facial deformity may develop
- **Erythrodontia** → teeth show **red fluorescence under UV light**

- Urine may appear **dark red / port-wine coloured**

Diagnosis

- Increased porphyrins in blood and urine
- **UV fluorescence** is useful.

Treatment

- Avoid sunlight and use strict **photoprotection**.
- Specialist treatment may include **red-cell suppression/transfusion strategies** or **hematopoietic stem-cell transplantation** in severe disease.

Remember

Congenital erythropoietic → **CHILD + SUN + RED**

5. Erythropoietic Protoporphyrria (EPP)

Cause

- **Ferrochelatase deficiency**
- ↓ Ferrochelatase → ↑ **free protoporphyrin in RBCs**

Clinical features

- Mainly **photosensitivity**
- Burning sensation and inflammatory reaction on sun exposure
- May develop cutaneous lesions
- Usually due to accumulation of **lipophilic protoporphyrin** in cell membranes.

Diagnosis

- ↑ free protoporphyrin in RBCs
- ↓ ferrochelatase activity
- RBCs show **red fluorescence at 620 nm**

Treatment

- **Sun protection**
- Avoidance of excessive sunlight
- Specialist therapy may include **afamelanotide** in appropriate patients.

6. Other Important Porphyrrias

ALA dehydratase deficiency porphyria

- Very rare.
- Deficiency of **ALA dehydratase (porphobilinogen synthase)**.
- Features resemble AIP:
 - **Abdominal pain**
 - **Neuropathy**
- ↑ **ALA** in plasma and urine.
- Differential diagnosis:
 - **Lead poisoning**
 - Tyrosinemia type I

Acquired porphyria — Lead poisoning

- Lead inhibits:
 - **ALA dehydratase**
 - **Ferrochelatase**
- ↓ heme → ↑ ALA synthase activity.
- Important distinction:

Acquired porphyria → **associated anemia**, unlike most congenital porphyrias.

7. Diagnosis of Porphyrias ★ ★ ★

Screening / detection

- **Urinary ALA and PBG** → important in acute neurovisceral porphyrias.
- **Porphyrins in urine, blood and feces** depending on the type.
- **Ehrlich's reagent** → detects porphyrin precursors.
- **UV fluorescence** → porphyrins show characteristic **red fluorescence**.
- Specific enzyme assay/genetic testing may confirm the diagnosis.

Key principle

Neurovisceral symptoms → look for **ALA + PBG**

Photosensitivity → look for porphyrins

8. Treatment — General Principles

1. Avoid precipitating factors

- Porphyrinogenic drugs
- Starvation
- Excess alcohol where relevant

2. Acute neurovisceral attacks

- High carbohydrate/glucose
- **Heme/hemin therapy** for significant attacks

3. Cutaneous porphyrias

- Strict **sun protection**
- Treat the specific porphyria appropriately.

The Table You Actually Need to Memorize

Porphyria	Enzyme defect	Main accumulation	Main presentation
AIP	↓ PBG deaminase	↑ ALA + PBG	 Abdominal + neurological , NO photosensitivity
Congenital erythropoietic	↓ Uroporphyrinogen III synthase	Type I porphyrins	 Severe photosensitivity , red urine, erythrodontia
EPP	↓ Ferrochelatase	↑ Protoporphyrin	 Photosensitivity/burning
ALAD deficiency	↓ ALA dehydratase	↑ ALA	 Abdominal pain + neuropathy
Lead poisoning	↓ ALAD + ferrochelatase	↑ precursors	 Porphyria + anemia

One-minute memory map

PORPHYRIA

→ Neurovisceral

- **AIP**
- Abdominal pain
- Neuropathy/psychiatric
- ↑ ALA + PBG
- **No photosensitivity**

→ Cutaneous

- **Congenital erythropoietic**
 - Severe photosensitivity

- Red urine
- Erythrodontia

- **EPP**

- Ferrochelatase ↓
- Protoporphyrin ↑
- Burning/photosensitivity

→ **Acquired**

- **Lead**
- ALAD + ferrochelatase inhibition
- **Anemia**